

EE 5239 Nonlinear Programming: Lecture 12

Nonnegative Matrix Factorization and Dictionary Learning

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Outline

- 1 Review
- 2 Nonnegative Matrix Factorization (NMF)
- 3 Mathematical Formulation
- 4 Algorithms for NMF
- 5 Dictionary Learning

This Time

- Introduce the Idea of **Nonnegative** Matrix Factorization (NMF)
 - ① Popular when nonnegative **factors** and **coefficients** are desired
 - ② Usually used to analyze the **nonnegative data**
- A classical algorithm called **Multiplicative Update**
- Introduce the notion of Dictionary Learning

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- **Question 1:** Create a set of “**basis images**”, or **factors**, that can be **linearly combined** to reconstruct the original (or new) images
- **Question 2:** Find **weights** to reproduce every input image from the basis images; one set of weights for each input image
- **What do I mean?**

Principal Component Analysis (PCA)

- A low-dimensional representation that minimizes the reconstruction error

$$\text{face}_i = \sum_{\ell=1}^L c_{i\ell} \times \text{eigenface}_{\ell}$$



“eigenfaces”

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- Any image can be reconstructed by the L eigen faces



“eigenfaces”

Principal Component Analysis (PCA)

- Given N images each of dimension K
- Construct the data matrix $V = [v_1, \dots, v_N] \in \mathbb{R}^{K \times N}$
- Find the “eigen faces”

$$V = W \Sigma G := WH,$$

- SVD, $W \in \mathbb{R}^{K \times L}$ contains the eigen faces



“eigenfaces”

PCA Weakness

- Is this really how humans characterize and identify faces?
- When viewing these as “**dictionary**”, do they make sense?



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- To construct a desired face, we **add up** some basis images (“+”) and **subtract** others (“-”)
- **Question:** What does it mean to subtract a face? How about a document?

Going from the whole to parts [Wachsmuth et al. 1994]

- Recording from neurons in the temporal lobe in the macaque monkey

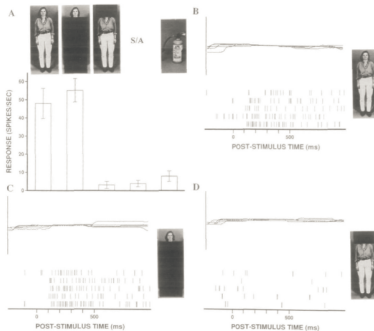


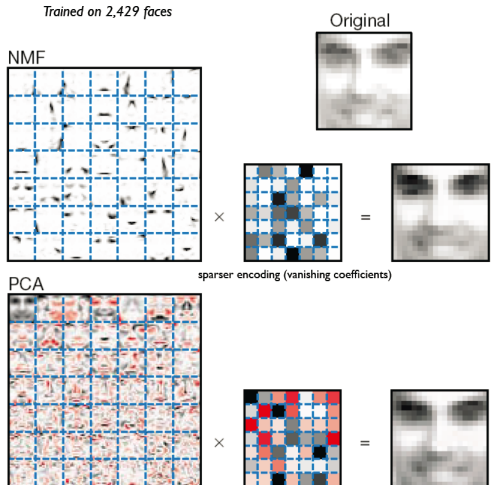
Figure 1. Examples of stimuli used for testing whole-body, head-only, and body-only stimuli in different views.

From the whole to parts [Wachsmuth et al. 1994]

- **Result:** Overall recorded from 53 neurons:
 - 17 (32%) responded to the head only
 - 5 (9%) responded to the body only
 - 22 (41%) responded to both the head and the body in isolation
 - 9 (17%) responded to the whole body only
- Suggests that **parts-based** representation is reasonable

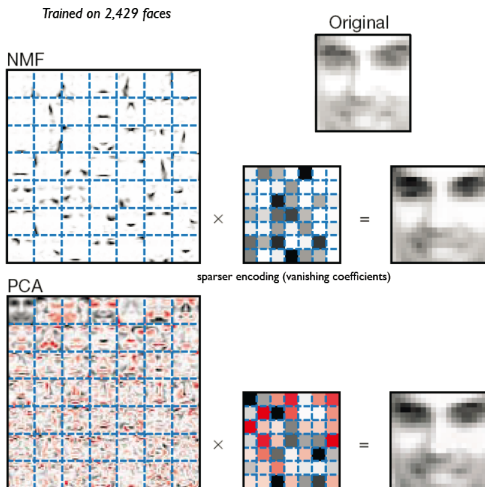
Nonnegative Matrix Factorization

- ① NMF is similar to PCA, but requires **nonnegative** combination of basis



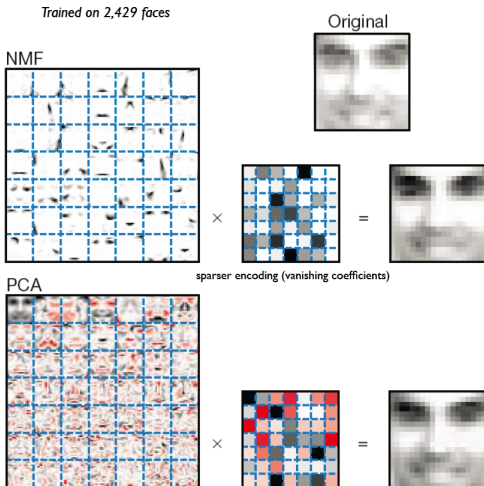
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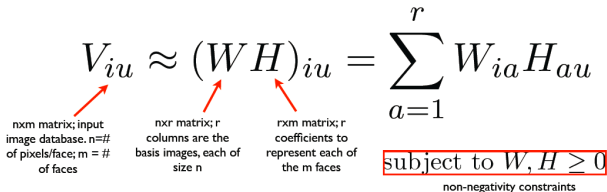
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- 3 **Several versions** of mouths, noises, etc
- 4 Better physical analogue



A Mathematical Formulation

- We would like to do the following

$$V_{iu} \approx (WH)_{iu} = \sum_{a=1}^r W_{ia} H_{au}$$



V : $n \times m$ matrix; input image database. $n = \#$ of pixels/face; $m = \#$ of faces

W : $n \times r$ matrix; r columns are the basis images, each of size n

H : $r \times m$ matrix; r coefficients to represent each of the m faces

subject to $W, H \geq 0$
non-negativity constraints

- The columns of V are the images (vectorized)
- The columns of W are the **basis images** (vectorized)
- The rows of H are the coefficients for nonnegative combination

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- What is an intuitive algorithm? First attempt, BCD:

$$W^{r+1} = \arg \min_W \|V - WH^r\|_F^2, \text{ subject to } W \geq 0$$

$$H^{r+1} = \arg \min_H \|V - W^{r+1}H\|_F^2, \text{ subject to } H \geq 0$$

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- Subproblem not easy to solve (nonnegative least squared, no closed-form solution)!

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 - ③ Treat each column/row of W or H as a block
- General algorithm fixing all the rest of variables, perform **projected gradient descent**
- The projection on nonnegative orthant is “easy”

A More Classical Algorithm: Multiplicative Update

- A classical algorithm by Lee and Seung (2000 NIPS, 9000+ citations so far) ¹
- Application in learning the parts of the objects (1999, Nature)²
- “...Here we demonstrate an algorithm for non-negative matrix factorization that is able to **learn parts of faces and semantic features of text**. This is in contrast to other methods, such as principal components analysis and vector quantization, that learn holistic, not parts-based, representations....” (1999, Nature)

A More Classical Algorithm: Multiplicative Update

- The algorithm can be simply written down as follows

$$[H^{r+1}]_{j,i} = [H^r]_{j,i} \frac{[(W^r)^T V]_{j,i}}{[(W^r)^T W^r H^r]_{j,i}}, \quad j = 1, \dots, K, \quad i = 1, \dots, N$$
$$[W^{r+1}]_{j,i} = [W^r]_{j,i} \frac{[V(H^{r+1})^T]_{j,i}}{[W^r H^{r+1} (H^{r+1})^T]_{j,i}}, \quad j = 1, \dots, M, \quad i = 1, \dots$$

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- If the first iterate $H^0 \geq 0$ and $W^0 \geq 0$, then the iterates **stay nonnegative**

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- If the first iterate $H^0 \geq 0$ and $W^0 \geq 0$, then the iterates **stay nonnegative**
- Also note that this is a **BCD-type of algorithm** (alternating between H-update and W-update)

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- As in the GD, construct a quadratic upper-bound

$$\begin{aligned} u_i(H_i, \{W^r, H^r\}) &:= f(H_i^r, \{W^r, H^r\}) + (H_i - H_i^r)^T \nabla_{H_i} f(H_i^r, \{W^r, H^r\}) \\ &\quad + \frac{1}{2} (H_i - H_i^r)^T \Phi_i(W^r, H_i^r) (H_i - H_i^r) \end{aligned}$$

where $\Phi_i(W^r, H_i^r)$ is a diagonal matrix given by

$$\Phi_i(W^r, H_i^r) := \text{Diag} \left(\frac{[(W^r)^T W^r H_i^r]_1}{[H_i^r]_1}, \dots, \frac{[(W^r)^T W^r H_i^r]_K}{[H_i^r]_K} \right)$$

Theory Behind Multiplicative Update

- Clearly, $\Phi_i(W^r, H_i^r) \succ 0$
- It is easy to show that $\Phi_i(W^r, H_i^r) \succ (W^r)^T W^r$, where $(W^r)^T W^r$ is the Hessian of the objective $f(\textcolor{red}{H}_i, \{W^r, H^r\})$
- What's happening graphically?

Theory Behind Multiplicative Update

- The following subproblem

$$H_i^{r+1} = \arg \min_{H_i \geq 0} u_i(H_i, \{W^r, H^r\})$$

has a unique solution given by

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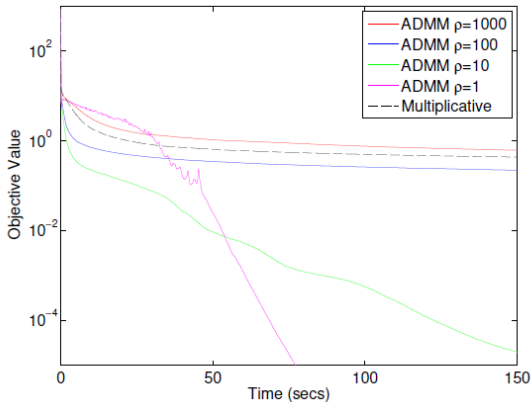
- The above update is the same as the multiplicative update rule
- Same reasoning applies for the W update

Multiplicative Update: Summary

- Related to the “Scaled Steepest Descent” method and BCD
- Each step in closed-form
- Nonnegativity is preserved due to
 - ① Good initial solution
 - ② The special structure of the iteration
- Convergence to KKT point of the NMF problem
- Numerically not always stable; can be slow

Numerical Comparison

- Numerically, the ADMM method (we will discuss this method later) seems to perform better (despite lack of convergence proof)



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- Put it into a matrix fom

$$\min_{\mathbf{D}, \mathbf{X}} \frac{1}{2} \|\mathbf{DX} - \mathbf{B}\|^2$$

Dictionary Learning (DL)

- If the observations are all nonnegative, we can have

$$\min_{\mathbf{D} \geq 0, \mathbf{X} \geq 0} \frac{1}{2} \|\mathbf{DX} - \mathbf{B}\|^2$$

- The DL approach is an **unsupervised learning approach**
- Learn the dictionary (basis) and the coefficient simultaneously
- Once the dictionary is learned, in the testing stage, receiving a “noisy” testing item $\hat{\mathbf{b}}_i$, we can perform the **reconstruction** by

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathbf{D}\mathbf{x} - \hat{\mathbf{b}}_i\|^2$$

- Note that the dictionary $\mathbf{D}$ is now fixed

Dictionary Learning (DL)

- Usually, in the training stage build a **large dictionary** (“over complete” dictionary)
- Think about the face reconstruction problem
- **B very large**, or **D very “fat”**
- Build **redundancy**, better testing performance
- Each training sample activates **a small number** of dictionary items

$$\min_{D, \{x_i\}} \frac{1}{2} \sum_{i=1}^N \|Dx_i - b_i\|^2 + \lambda \|x_i\|_1$$

- **Dictionary Learning + Sparse Coding**
- Testing Stage, solve

$$\min_x \frac{1}{2} \|Dx - \hat{b}_i\|^2 + \lambda \|x\|_1$$